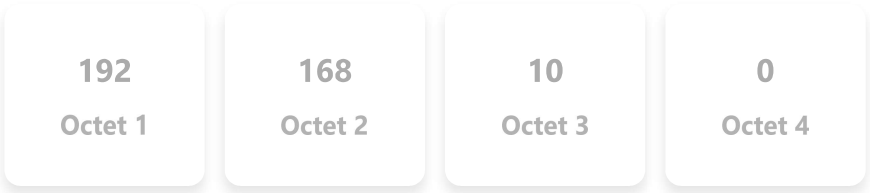
🌐 Subnetting Visual Guide

Master subnetting with clear visual representations

1. IP Address Structure

An IPv4 address consists of 4 octets (32 bits total), displayed in dotted decimal notation:



Each octet = 8 bits | Total = 32 bits

Binary Representation:

```
11000000.10101000.00001010.00000000
```

2. Network vs Host Portions

The subnet mask determines which bits represent the network and which represent hosts:

Example: 192.168.10.0/24

```
IP Address: 11000000.10101000.00001010.00000000
Subnet Mask: 11111111.11111111.11111111.00000000
```

```
Subnet Mask: 255.255.255.0
```

Network Portion (24 bits) Host Portion (8 bits)

3. Borrowing Bits for Subnetting

When we subnet 192.168.10.0/24 into 4 subnets, we borrow 2 bits from the host portion:

Original /24 Network:

1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0

After Borrowing 2 Bits → /26 Network:

1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0

Original Network Bits Borrowed Bits (New Subnet IDs)

Remaining Host Bits

Number of Subnets = 2^n where n = borrowed bits

Number of Subnets = 2^2 = **4 subnets**

Usable Hosts per Subnet = 2^h - 2 where h = remaining host bits

Usable Hosts = 2^6 - 2 = **62 hosts**

4. The Four Subnets Created

Subnet 1 - Sales Department

Network Address:
192.168.10.0/26

First Host: 192.168.10.1

Last Host:
192.168.10.62

Broadcast:
192.168.10.63

Usable Hosts: 62

```
Range: .00000000 to
.00001111
Decimal: .0 to .63
```

Subnet 2 - Engineering

Network Address:
192.168.10.64/26

First Host:
192.168.10.65

Last Host:
192.168.10.126

Broadcast:
192.168.10.127

Usable Hosts: 62

```
Range: .01010000 to
.01011111
Decimal: .64 to .127
```

Subnet 3 - Marketing

Network Address:
192.168.10.128/26

First Host:
192.168.10.129

Subnet 4 - HR Department

Network Address:
192.168.10.192/26

First Host:
192.168.10.193

Last Host:
192.168.10.190

Broadcast:
192.168.10.191

Usable Hosts: 62

```
Range: .10100000 to
.10101111
Decimal: .128 to .191
```

Last Host:
192.168.10.254

Broadcast:
192.168.10.255

Usable Hosts: 62

```
Range: .11110000 to
.11111111
Decimal: .192 to .255
```

5. Complete Subnet Masks Reference (/8 to /32)

CIDR	Subnet Mask	Usable Hosts	Common Use
/8	255.0.0.0	16,777,214	Class A network
/9	255.128.0.0	8,388,606	Large enterprise
/10	255.192.0.0	4,194,302	Large enterprise
/11	255.224.0.0	2,097,150	Large enterprise
/12	255.240.0.0	1,048,574	Large enterprise
/13	255.248.0.0	524,286	ISP allocation
/14	255.252.0.0	262,142	ISP allocation

CIDR	Subnet Mask	Usable Hosts	Common Use
/15	255.254.0.0	131,070	ISP allocation
/16	255.255.0.0	65,534	Class B network
/17	255.255.128.0	32,766	Medium enterprise
/18	255.255.192.0	16,382	Medium enterprise
/19	255.255.224.0	8,190	Medium enterprise
/20	255.255.240.0	4,094	Small enterprise
/21	255.255.248.0	2,046	Small enterprise
/22	255.255.252.0	1,022	Small business
/23	255.255.254.0	510	Small business
/24	255.255.255.0	254	Class C / Standard LAN
/25	255.255.255.128	126	Large department
/26	255.255.255.192	62	Medium department
/27	255.255.255.224	30	Small department
/28	255.255.255.240	14	Small office/VLAN
/29	255.255.255.248	6	Tiny subnet
/30	255.255.255.252	2	Point-to-point links
/31	255.255.255.254	0 (2 special)	Point-to-point (RFC 3021)
/32	255.255.255.255	0 (1 host)	Single host/loopback

Key Takeaways

✓ More borrowed bits = More subnets but fewer hosts per subnet

✓ Fewer borrowed bits = Fewer subnets but more hosts per subnet

✓ Always subtract 2 from total hosts (network + broadcast addresses)

✓ Subnet increment = value of the last borrowed bit position